Paravirt Scheduling:
Framework for better CPU usage

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Agenda

- Problem Statement and proposed Solution
- Existing methods
- Journey so far
- Design goals
- Implementation details
- s390 background
- Performance numbers
- Open questions

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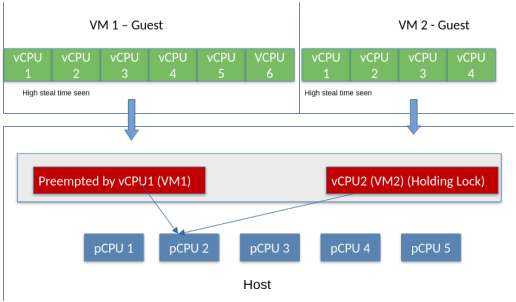
Problem Statement: vCPU Overcommit and vCPU Preemption

- Multiple VMs sharing the underlying CPU resources, with overcommit (sum of vCPUs > pCPUs)
- High CPU utilization in multiple VMs at the same time leads to vCPU preemption
- Why vCPU preemption can be expensive:
  - **vCPU context:** a vCPU may be holding a mutex or spinlock
  - **Hypervisor overhead:** hypervisors may schedule at a different granularity than the guest, introducing extra delays and cache/TLB thrashing on vCPU migration
- **Design constraint:** the hypervisor is non-cooperative – no host-side changes, no inter-guest channel (e.g. PR/SM, PowerVM)



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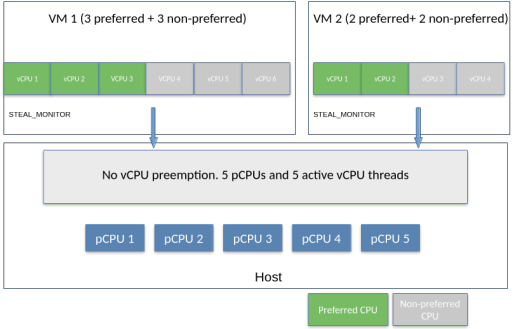
Proposed Solution: steal-driven vCPU backoff

- The vCPUs each guest can safely use are denoted as *preferred* CPUs
- **Aim:** sum of preferred CPUs $\leq$ pCPUs, which achieves less vCPU preemption
- **Mechanism:** each guest independently adjusts its preferred CPU set from locally observed steal time
- **Fairness:** all guests run the same backoff logic, so they converge without talking to each other



Advantage of a context switch within the vCPU instead:

- Lockholder runs to completion (`preempt_disable()`)
- No hypervisor overhead

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Existing Methods

Discussed in detail at LPC 2025 – why no existing methods work. A quick glance:

- **Explicit affinity:** can't expect a user to set affinity for each workload
- **CPU hotplug:** slow, expensive, breaks user-specified affinities
- **Isolated cpuset partitions:** needs sched domain rebuild, breaks user-specified affinities
- **Capacity-aware scheduling:** doesn't push tasks from a CPU under high load

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Journey so far...

- **LPC 2024** – introduced the problem statement. A new load balancing type `cpu_parked` was used. This was a PoC, slow in moving tasks since it depended on the load balancer and heavily on stats.
  [RFC PATCH 0/2] sched/fair: introduce new scheduler group type `group_parked`
- **LPC 2025:** a few revisions later, switched to a push-task-mechanism-based method. Much faster and more policy-driven than stats.
  [PATCH 00/17] Paravirt CPUs and push task for less vCPU preemption

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Design Goals

- Keep runnable vCPUs close to pCPU count under contention
- Preserve user-specified CPU affinity
- Keep the solution scheduler-native and lightweight
- Provide a debug tunable to adjust based on the platform
- Expose the new CPU state information for consumption
- Do not preclude future hypervisor-assisted solutions

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New CPU State: `cpu_preferred_mask`

- Represents CPUs the scheduler should currently prefer
- Not a hard exclusion: affinity is still respected
- Dynamic and policy-driven
- Maintains the design construct:
  preferred $\subseteq$ online $\subseteq$ possible

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Sched feature: `STEAL_MONITOR`

- In `sched_tick()`, check if it is time to compute steal time
- Kick off a work function to compute steal time – keeps `sched_tick()` latency minimally affected
- Compute steal time ratio from `CPUTIME_STEAL` across all online CPUs
- If steal time is above the threshold, reduce preferred CPUs
  - Current implementation reduces by one core, because some hypervisors (e.g. PowerVM) schedule at core granularity – this yields the best benefit
- If steal time is below the threshold, increase preferred CPUs
- Simple heuristics avoid oscillations: only consecutive high/low steal values are considered

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Core APIs being updated

- `is_cpu_allowed()`
  - Try to select a preferred CPU
  - If task affinity is only on non-preferred CPUs, return a non-preferred CPU
  - Ensure overhead is minimal for the majority use case
  - Used by `select_fallback_rq()`, the push-task mechanism, and wake-ups
- `available_idle_cpu()`
  - Make the CPU available if it is part of preferred CPUs
  - Handles wake-up decisions; used by CFS
- `__cpupri_find()`
  - Consider only preferred CPUs when finding `lowest_rq` for RT tasks

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How it all fits together

- On task wake-up, when choosing a CPU, pick a preferred CPU where possible
- Load-balance only among preferred CPUs, so load balance and push-task don't fight each other; load stays on preferred CPUs
- At `sched_tick()`, the first check: is this CPU non-preferred? If so, push the task out via a stopper thread. Only the current task is pushed (keeps it simple). Since the work is not done in `sched_tick()`, IRQ latency shouldn't increase much
- A task is pushed only if it doesn't break affinity. If a task is bound only on non-preferred CPUs, there is no point pushing it out

[PATCH v2 00/17] sched/paravirt: Introduce `cpu_preferred_mask` and steal-driven vCPU backoff

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User Visibility & Control

- `/sys/devices/system/cpu/preferred` – current preferred CPUs
- `STEAL_MONITOR` sched feature
- Debugfs knobs for thresholds and sampling period – escape hatch for debugging, not a per-workload admin interface
- Plan: pick per-architecture defaults

```
# echo STEAL_MONITOR > /sys/kernel/debug/sched/features
# echo NO_STEAL_MONITOR > /sys/kernel/debug/sched/features

# pwd
/sys/kernel/debug/sched
steal_mon_high:500
steal_mon_low:200
steal_mon_period:1000
```


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s390 challenges

- Large system: ~200 cores
- There is no bare-metal Linux on s390
- Proprietary type-1 hypervisor: PR/SM
  - Unlikely to add Linux-specific enhancements
- Overcommit and steal time
- Whole-system performance vs. single-VM performance

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s390 Hiperdispatch

- PR/SM designates CPUs as high, medium, and low, and lets the guest know which is which
- PR/SM frequently steals low CPUs for a long time
- Problem: exits to PR/SM are expensive
- Idea: avoid scheduling threads on low CPUs that had steal time recently

Hiperdispatch

- Created by Tobias Huschle and Mete Durlu
- Upstream since v6.12
- Uses capacity-aware scheduling
- Does not handle heavy overcommit

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s390 Warning Track Interrupt

- Problem: contention if a lock holder is stolen
- PR/SM sends an interrupt shortly before stealing a CPU
- Idea: park CPUs that are about to be stolen to prevent them from taking locks

WTI

- Created by Tobias Huschle and Mete Durlu
- Upstream since v6.12
- Schedules a real-time thread to yield to the hypervisor ASAP

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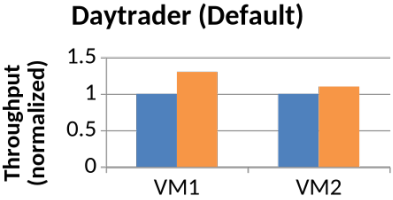
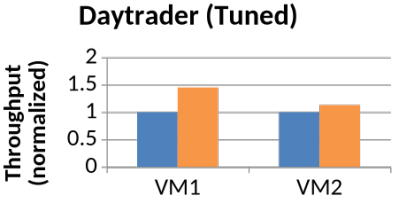
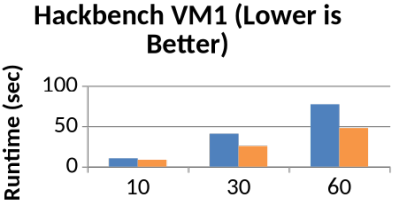
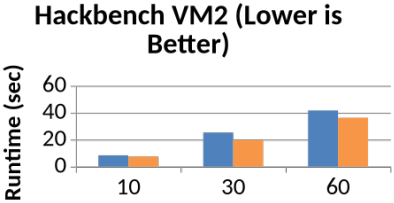
Performance numbers: PowerPC / PowerVM hypervisor









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Performance numbers: s390 PR/SM

- s390 host with 37 cores
- 4 partitions with 37 shared cores each
- Workload 1: `stress-ng --matrix 37`
- Workload 2: `uperf` ping-pong
- Workload 3: `uperf` streaming

Results:

- ~2x throughput improvement for all workloads
- Steal time reduction from ~60% to ~5%

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Performance numbers: KVM

- 2–16 VMs
- 4–32 CPUs each
- s390 host with 16 cores
- x86_64 host with 32 cores
- Up to 4x overcommit
- Each VM runs the same benchmark
  - `hackbench`, `pgbench`, `schbench`, `sysbench`
- Throughputs are summed across all VMs
- Repeated 10 times to quantify noise

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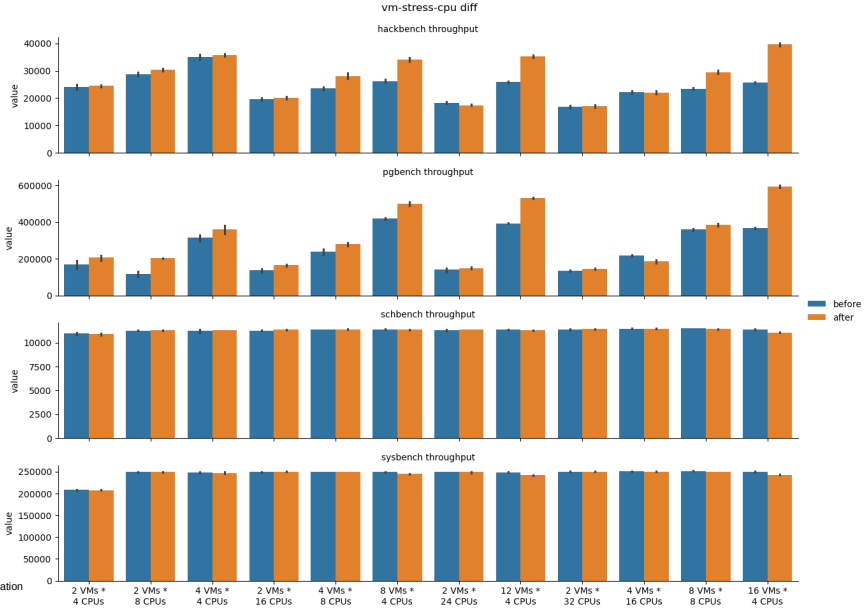
Performance numbers: s390 KVM



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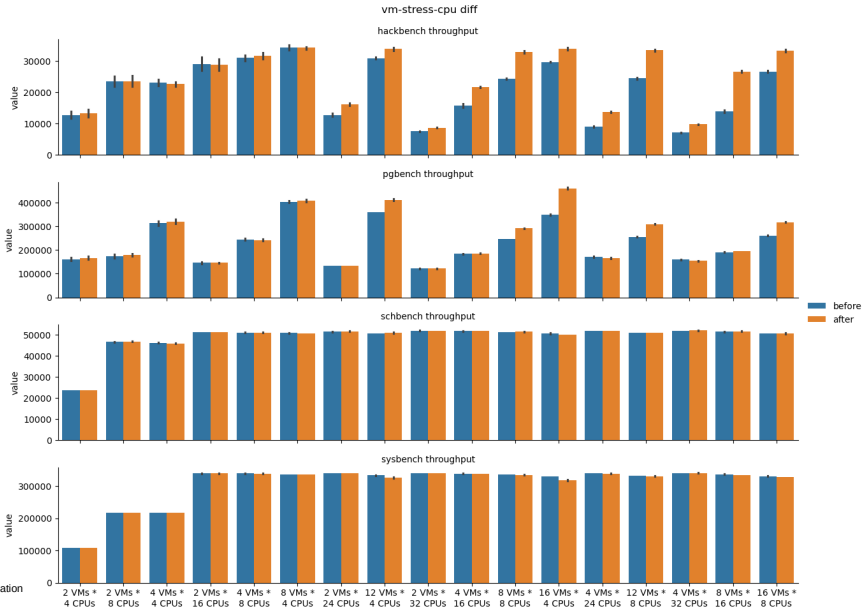
Performance numbers: x86_64 KVM



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Open Questions: Concept

- Is the design sound for the majority use case?
- Upstream readiness?
- Is there any !SMT system interested in this?
- How to avoid oscillations in complex overcommit cases?
- `irqbalance` must be made preferred-mask-aware; when it is off, should the kernel push IRQs off non-preferred CPUs?
- Should `sched_ext` tasks be pushed too, in addition to the fair and RT tasks handled today?

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Open Questions: Implementation

- Arch-specific hook to choose which CPU to drop from the preferred mask, e.g. based on power entitlements or s390 polarization
  - Default: NUMA-aware – pack onto one node, or spread across nodes? Spreading likely wins – more resources overall at the cost of cross-node penalties
- `STEAL_MONITOR` off by default, with each arch opting in – is that the right default?

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Backup: avoiding host under-utilization

- **Implemented today:** fast shrink, slow grow – same as TCP
- **Idea:** EWMA-filter the steal time signal – same as PELT
- **Idea:** rate limit preferred-mask changes
- **Idea:** randomize the sampling phase and mask-change rate across guests to avoid synchronized decisions

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Backup: fairness without coordination

- Non-cooperative hypervisors are the main reason this work exists
- Coordination via the host is not an option on PR/SM or PowerVM
- Similar to TCP: no central coordinator
- One could do better there with explicit host cooperation, but the uncoordinated mechanism already improves things

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Backup: is this SMT-specific?

- No. Nothing in the mechanism depends on SMT
- Current implementation adjusts the preferred mask by one *core* only because some hypervisors (e.g. PowerVM) schedule at core granularity

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Backup: steal time collection context

- Under heavy overcommit the kworker itself can be stolen
- **Idea:** keep the computation inside the tick
- Precedent: PELT already runs inline from the tick, so this would not be worse: `sched_tick()` → `task_tick_fair()` → `entity_tick()` → `update_load_avg()` → `__update_load_avg_se()`

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Backup: relative steal time

- Many VMs with low utilization can cause overcommit
- Absolute steal time misses this
- **Idea:** divide steal time by non-idle time

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Backup: replacing `stop_one_cpu_nowait()`

- Current v2 pushes tasks off non-preferred CPUs via the CPU stopper – heavy-handed (runs at `stop_sched_class`, above RT/DL) for what is a soft placement hint
- **Idea:** build on Vincent Guittot's CFS push-task mechanism
- [PATCH 0/6 v8] sched/fair: Add push task mechanism and handle more EAS cases
- **Idea:** extend it to push all tasks off a CPU
- **Idea:** reuse the CPU-offline task pushing code

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